(Please write your Enrollment Number) Enrollment No._____

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name:<BTech CSE-AI>, Semester:<VII>
(November-December, 2023)

Subject Code: BAI 403	Subject: Computer Vision
Time : 3 Hours	Maximum Marks :60

Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(2.5*8=20)	
	(a) How is supervised learning different from unsupervised learning Explain with an example		
	(b) What is Principal Component Analysis		
	(c) Explain the design cycle of a pattern recognition system.		
	(d)What are the different types of discriminant functions? Just name them.		
	(e) Name the stages of computer vision and their use in understanding an image.		
	(f) Why Max-Pooling makes the Convolutional Neural Network translation invariant (for small translations of the input)?		
	(g) What is the use of fully convolutional network in CV?		
	(h) For max-pooling done on a 6x6x3 image with a filter of size f=2 and padding p=0 with stride s=1, what would be the size of the output image?		
	UNIT-I		
Q2	What is Computer Vision? Why is vision so difficult? Provide six real-world examples of computer vision and explain.	(10)	
Q3	What is the optimization function of the loss function? Explain its role in CV.	(10)	
	UNIT-II		
Q4	Explain any two examples of CNN architectures.	(10)	
Q5	Explain K-means Clustering for Vector Quantization in detail.	(10)	
	UNIT-III		
Q6	Make a comparative analysis between CNN and RNN and their implementation in computer vision.	(10)	
Q7	What is a Vision Transformer for image prediction? Explain it with an application	(10)	
	UNIT-IV		
Q8	What is data augmentation technique in computer vision? What is its use?	(10)	
Q9	Explain the use of GAN in computer vision?	(10)	